Multimodal Sensor Fusion for Human Activity Recognition: A Comparative Study of Early vs. Late Fusion in Deep Learning Architectures.

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Abstract— **Human Activity Recognition (HAR) using wearable sensors is central to applications ranging from health monitoring to smart environments. However, developing models that generalize across different users remains challenging due to individual variations in movement patterns and hardware-specific noise. This study evaluates deep learning approaches for multi-sensor HAR using the CogAge dataset. We analyze 27 inertial sensor axes collected from a smartphone, smartwatch, and smart glasses. To distinguish between spatial movement features and temporal dynamics, we compare a baseline Long Short-Term Memory (LSTM) network against a hybrid Convolutional Neural Network-LSTM (CNN-LSTM). We also assess two multimodal integration techniques: early fusion and late fusion. Following synchronization to 50 Hz and segmentation into 128-timestep sliding windows, our evaluations demonstrate that late fusion combined with a CNN-LSTM architecture best isolates localized sensor noise. Under a strict leave-one-subject-out evaluation protocol, this configuration achieved a subject-independent accuracy of 55.18% across 12 atomic activities, outperforming baseline approaches.**

Index Terms—Human Activity Recognition, Sensor Fusion, Deep Learning, CNN-LSTM, Wearable Sensors.

I. INTRODUCTION

The proliferation of wearable technology has made Human Activity Recognition (HAR) an active area of research, particularly for applications in healthcare, fitness tracking, and ambient assisted living. The core objective of HAR is to translate continuous, multivariate inertial sensor data into discrete activity classifications.

While recent advances in deep learning have improved classification accuracy, building models that generalize to unseen users remains a primary bottleneck. Inertial data is inherently noisy, and variations in device placement—such as a smartphone shifting in a pocket—add inconsistency. Furthermore, HAR systems are highly susceptible to intra-class variability; different individuals perform the exact same activity with distinct postures, gait speeds, and biomechanical rhythms. This makes cross-subject generalization one of the most persistent challenges in pattern recognition.

[1]To address these challenges, this paper investigates the efficacy of hybrid deep learning architectures and sensor fusion strategies using the multi-device CogAge dataset[1]. Because human movement involves both immediate physical impacts (spatial features) and rhythmic sequences (temporal features), we benchmark a standard LSTM network against a CNN-LSTM model designed to capture both dimensions. Additionally, because aggregating data from the wrist, head, and pocket can amplify device-specific noise (such as magnetic interference from a smartphone compass), we evaluate the trade-offs between early (data-level) and late (feature-level) fusion.

The specific contributions of this work are as follows:

1) We establish a preprocessing pipeline to synchronize and segment 27 independent inertial axes from three distinct wearable devices into unified sequence matrices.

2) We benchmark CNN-LSTM architectures against pure RNN approaches, demonstrating the utility of convolutional layers in filtering high-frequency hardware noise prior to sequential modeling.

3) We show through rigorous leave-one-subject-out cross-validation that feature-level late fusion prevents localized sensor anomalies from degrading overall model performance, leading to better generalization on unseen subjects

II. METHODOLOGY

This study investigates deep learning architectures and sensor fusion strategies for recognizing human activities using multimodal sensor data. To systematically evaluate these approaches, we structured our methodology into data preprocessing, architectural design, and fusion strategy implementation.

A. Dataset Description

We utilize the multimodal CogAge dataset, which captures inertial data from three distinct wearable devices: a smartphone (pocket), a smartwatch (wrist), and smart glasses (head). These devices collectively provide 27 sensor axes. The smartphone contributes 15 axes (3-axis combinations of the accelerometer, gyroscope, gravity sensor, linear acceleration, and magnetometer), while the smartwatch and smart glasses each contribute 6 axes (3-axis accelerometer and 3-axis gyroscope). The dataset encompasses recordings of 12 distinct daily human activities, ranging from static postures (e.g., Sitting, Standing) to dynamic, complex motions (e.g., Walking, Squatting, DryOffHands).

B. Data Preprocessing and Segmentation

To address hardware desynchronization across the three independent device clocks, all sensor streams were linearly interpolated and synchronized to a uniform sampling frequency of 50 Hz.

To capture the temporal dynamics of the inertial signals, the continuous multivariate time-series data was partitioned using a sliding window technique. Let the synchronized continuous signal be denoted as $S \in \mathbb{R}^{T \times F}$, where T is the total number of timesteps and $F = 27$ represents the full spatial feature set. We applied a fixed-length window $W = 128$ (equivalent to 2.56 seconds of movement) with an overlap step size of $O = 64$ (a 50% overlap). This transformation maps the continuous stream into a discrete sequence tensor $X \in \mathbb{R}^{N \times 128 \times 27}$, where N is the total number of extracted sequence samples. Finally, Z-score normalization was applied to standardize the amplitude of the sensor readings across all modalities.

C. Deep Learning Architectures

Two distinct recurrent models were implemented to classify the extracted time-series windows.

1) **Pure LSTM (Baseline):** [3]The first architecture is a standard Long Short-Term Memory network. This model receives the raw, unrefined sequence matrices and relies entirely on its internal gating mechanisms to capture temporal relationships and filter noise in the sequential sensor data.[3]

2) **CNN-LSTM (Proposed):** The second model is a hybrid architecture utilizing 1D Convolutional Neural Networks (Conv1D) preceding the recurrent layers. The convolutional layers function as localized feature extractors, identifying sharp spatial patterns (e.g., impact spikes from footsteps) before the LSTM layers model the temporal dependencies of those patterns over time.

D. Sensor Fusion Strategies

To determine the optimal method for integrating data across the three body locations, we compared two distinct sensor fusion strategies.

1) **Early Fusion (Data-Level):** In this approach, the sensor data is concatenated into a single input matrix before model training. The neural network receives the complete tensor $X \in \mathbb{R}^{128 \times 27}$ at the input layer, forcing the initial network weights to simultaneously process the clean data from the wrist and head alongside the noisy magnetic data from the pocket.

2) **Late Fusion (Feature-Level):** This strategy processes each device in separate neural network branches. The input tensor is sliced into hardware-specific sub-tensors: $X_{phone} \in \mathbb{R}^{128 \times 15}$, $X_{watch} \in \mathbb{R}^{128 \times 6}$, and $X_{glasses} \in \mathbb{R}^{128 \times 6}$. Each branch applies its own spatial and sequential modeling. The high-level feature representations from these isolated branches are then concatenated in later layers immediately prior to the final dense classification layer.

III. EXPERIMENTAL SETUP

To rigorously evaluate the generalization capabilities of the proposed architectures, performance was measured using classification accuracy and categorical cross-entropy loss. Experiments were conducted under two distinct cross-validation protocols to isolate the effects of intra-class variability.

A. Subject-Dependent Evaluation

A 5-Fold Cross-Validation strategy was employed to establish a baseline for model capacity. The complete dataset, containing samples from all subjects, was randomly shuffled and partitioned into five equal folds. Four folds were utilized for training and one for testing, iterating until every fold served as a test set. This setting evaluates the network's ability to classify activities when it has been explicitly exposed to the unique biomechanical signatures of the users during the training phase.

B. Subject-Independent Evaluation

To assess true real-world applicability, a Leave-One-Subject-Out (LOSO) cross-validation protocol was implemented. The models were trained entirely on the data of all subjects except one, and subsequently evaluated strictly on the unseen subject. This protocol prevents the network from memorizing subject-specific quirks, forcing it to rely purely on the underlying, universal physics of the 12 atomic activities.

IV. RESULTS AND DISCUSSION

The experimental results systematically compare the Pure LSTM and CNN-LSTM architectures under both Early and Late Fusion paradigms. The outcomes highlight the critical role of feature isolation in multimodal wearable systems.

TABLE I : **PERFORMANCE COMPARISON OF HAR ARCHITECTURES**

Model Architecture	**Fusion Strategy**	**Subject-Dependent (5-Fold CV)**	**Subject-Independent (LOSO)**
CNN-LSTM	Late Fusion	95.59%	55.18%
Pure LSTM	Late Fusion	87.37%	47.71%
CNN-LSTM	Early Fusion	96.01%	45.18%
Pure LSTM	Early Fusion	91.00%	46.14%

A. The Late Fusion Advantage and Sensor Noise

The Late Fusion CNN-LSTM configuration achieved the highest subject-independent accuracy of 55.18%. Conversely, when the CNN-LSTM utilized Early Fusion, the subject-independent accuracy degraded significantly to 45.18%. This performance gap is directly attributable to the management of localized sensor noise.

In Early Fusion, the network receives a concatenated 27-feature input vector. Because the smartphone contributes 15 of these axes—including highly volatile magnetometer readings subject to indoor ferromagnetic interference—the unified network struggles to disentangle this noise from the clean kinematic data provided by the smartwatch and smart glasses. Consequently, the Early Fusion model overfits to the environmental and subject-specific noise of the training data (achieving a peak 96.01% dependent accuracy) but fails to generalize.

Late Fusion effectively acts as a structural regularizer. By processing the 15-axis smartphone data in an isolated branch, the network bounds the magnetic and positional noise, preventing it from corrupting the spatial-temporal feature extraction of the wrist and head kinematics until the final representation layers.

B. Impact of Spatial Feature Extraction

[2],[4]Consistent with contemporary HAR literature, the hybrid CNN-LSTM consistently outperformed the Pure LSTM baseline across Late Fusion configurations. The Conv1D layers successfully operated as automated, high-frequency noise filters[2],[4]. By extracting localized spatial features—such as the sharp acceleration spike of a heel strike—the convolutional layers provided a refined, lower-dimensional feature map. This allowed the subsequent LSTM layers to focus exclusively on modeling the temporal evolution of the activity, rather than expending internal capacity filtering raw inertial noise.

C. The Generalization Challenge

The substantial disparity between subject-dependent (~95%) and subject-independent (~55%) accuracies quantitatively illustrates the severity of intra-class variability in human kinetics. While the network easily maps high-dimensional patterns to specific users, predicting the gait, posture, and execution speed of an unseen human introduces massive statistical variance. Nevertheless, the Late Fusion CNN-LSTM's 55.18% accuracy across 12 highly complex classes represents a greater than six-fold improvement over the random-chance baseline (8.33%). This confirms that the multimodal network successfully abstracted the core spatial-temporal concepts of the activities, rather than merely memorizing the training distribution.

V. CONCLUSION

This paper presented a systematic evaluation of deep learning architectures and multimodal sensor fusion strategies for Human Activity Recognition using the CogAge dataset. Recognizing daily human activities across different individuals remains a complex sequence-modeling challenge due to severe intra-class variability and heterogeneous hardware noise. To address these challenges, we benchmarked Pure LSTM networks against hybrid CNN-LSTM architectures under both early and late fusion paradigms.

Our findings conclusively demonstrate that the integration of 1D Convolutional layers significantly enhances sequence modeling by extracting robust spatial features prior to temporal analysis. Furthermore, the experiments revealed the critical importance of feature-level isolation when processing multimodal data. While Early Fusion led to severe overfitting by blending noisy smartphone magnetometer data with clean kinematic signals, Late Fusion effectively acted as a structural regularizer. By processing the distinct hardware modalities in isolated branches, the Late Fusion CNN-LSTM architecture mitigated localized sensor noise and achieved the highest generalization capability, culminating in a peak subject-independent accuracy of 55.18% across 12 complex atomic activities. Ultimately, this study confirms that a Late Fusion CNN-LSTM provides the optimal balance of spatial-temporal feature extraction and noise isolation for multi-device wearable HAR systems.

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